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REVISION R5 · Days 25–30: week notes and a 30-question test

UNDERSTAND IT FIRST

How to use this page (about 3 hours): (1) Blurt first (30 min): before reading anything, take a blank page and write all you remember for each day's three topics. **(2) Check (30 min):** compare with the week notes below; mark what you missed in **red**. Only the red points go into your notebook. **(3) Test (60 min):** take the 30 questions without notes, then read every explanation, including those you got right. **(4) Fix (30 min):** for each wrong answer, write one line in the 'My mistakes' box at the bottom. **Next revision of this week:** in 7 days, then 30 days.

Week notes: write these in your notebook

Geography (Days 25–30)

Day	Points to write
25: Earth in the Solar System	Light year (distance); AU ~150 million km; Sun 99.8% mass; planets in order with one fact each (Venus hottest and retrograde, Jupiter largest, Saturn least dense, Uranus 98° tilt); Pluto dwarf 2006; perihelion 3 Jan, aphelion 4 July; tilt 23.5°; solstices 21 June, 22 Dec; equinoxes 21 Mar, 23 Sept; 15° = 1 h; IST 82.5° E; solar eclipse new moon, lunar full moon.
26: Interior of the Earth	P (all media, fastest), S (solids only), surface waves (most damage); shadow zone 105°–145°; Moho, Gutenberg (2,900 km), Lehmann (5,150 km), Conrad; outer core liquid; crust O > Si > Al > Fe; three rock types and changes; plate boundaries; Richter vs Mercalli; zones II–V; Barren Island active.
27: Landforms	Endogenic vs exogenic; fold, horst, graben, volcanic, residual mountains; plateau types; river stages (V-valley → meander → oxbow lake, delta); estuary (Narmada, Tapi); wind (barchan, mushroom rock), glacier (U-valley, cirque, fjord, moraine), waves (stack, spit, lagoon), karst (stalactite, stalagmite; Borra, Belum).
28: Atmosphere	N_2 78%, O_2 21%, Ar 0.93%, CO_2 0.04%; troposphere (weather, 6.5°C/km; thickest at equator), stratosphere (ozone), mesosphere (coldest), ionosphere (radio); albedo ~30%; conduction, convection, advection; inversion; MSL 1013 mb; seven pressure belts.
29: Winds and climate	Coriolis (right in N, zero at equator); trades, westerlies (roaring forties), polar easterlies; land and sea breezes; loo, mango showers, Kal Baisakhi, western disturbances; Chinook, Sirocco, Harmattan, Mistral; cyclones (anticlockwise in N; Hudhud, Titli, Michaung); three rain types; rain shadow; Köppen and India's types (As = Coromandel).
30: Oceans	Floor relief; salinity 35‰ (Bay of Bengal < Arabian Sea); tides (Moon; spring at new/full moon, neap at quarters; Fundy); warm (Gulf Stream, Kuroshio, Agulhas, Brazil) and cold

	(Labrador, Humboldt, Benguela, Canary, California, Oyashio) currents; Grand Banks; El Niño weakens monsoon; positive IOD helps; coral reef types.

Polity and Economy (Days 25–30)

Day	Points to write
25: Panchayati Raj	Balwant Rai Mehta 1957 (three tiers; samiti executive); Rajasthan and AP 1959; Ashok Mehta 1978 (two tiers; parties); G.V.K. Rao 1985; Singhvi 1986; 73rd: in force 24 April 1993; Part IX, 243–243-O; 11th Schedule (29); Gram Sabha 243A; women ≥1/3; 5 years, 6 months; SFC, SEC; PESA 1996; AP Act 1994 (GP, MPP, ZPP).
26: 74th Amendment	Madras 1688; Ripon 1882; in force 1 June 1993; Part IXA, 243P–243ZG; 12th Schedule (18); Nagar Panchayat, Council, Corporation; Wards Committees (3 lakh+); DPC 243ZD; MPC 243ZE (10 lakh+); Cantonment Boards (Defence); problems; Smart Cities (AP: 4), AMRUT, SBM-U.
27: National income	GDP → GNP (+NFIA) → NNP (−depreciation) → NI (−net indirect taxes); GVA at basic prices; nominal vs real, deflator; product, income, expenditure methods; included and excluded items; Naoroji, V.K.R.V. Rao, Mahalanobis 1949, NSO 2019; base year 2011–12 → 2022–23.
28: Occupations, sectors	Three sectors; Clark–Fisher; Lewis; GVA ~18/28/54 vs jobs ~46/25/29; PLFS (NSO), LFPR, WPR, UR; usual status (365 days) vs CWS (7 days); disguised, seasonal, structural, frictional; ~90% informal; e-Shram; Labour Codes.
29: Growth vs development	Quantitative vs qualitative; Sen 1999; Brundtland 1987; PQLI (Morris 1979); HDI (1990; India ~130); MPI (UNDP–OPHI; NITI: 24.8 crore out); SDGs (17, 169); SDG India Index; Harrod–Domar, Rostow, Nurkse, Kuznets; Hindu rate; Swarna Andhra @2047.
30: Planning, NITI	Visvesvaraya 1934, NPC 1938, Bombay Plan 1944; Planning Commission 15 Mar 1950; NDC 1952; 12 Plans (1st agriculture, 2nd Mahalanobis, 3rd failed, holidays 1966–69, 5th ended 1978, 8th indicative, 11th inclusive, 12th last); NITI 1 Jan 2015 (Governing Council, indices, no funds, Aspirational Districts).

Science and environment (Days 25–30)

- **Ecology (25):** Haeckel, Tansley; food chains; 10% law; pyramids (numbers inverted in a tree, biomass in the ocean, energy always upright); N-cycle organisms; succession (lichens pioneer); interactions; biomagnification, eutrophication. **Biodiversity (26):** three levels; India 2.4% land, 7–8% species; hotspot criteria and India's four (Eastern Ghats not one); HIPPO; in-situ vs ex-situ; CBD, Cartagena, Nagoya, Kunming–Montreal (30×30); BD Act 2002 (NBA, SBB, BMC).
- **Wildlife (27, 28):** IUCN categories; CITES appendices; CMS, Ramsar; 42nd Amendment; WPA 1972 (2022: four schedules); NTCA, WCCB; FCA 1980 → 2023 Adhiniyam; FRA 2006; CAMPA 2016; Project Tiger 1973, Elephant 1992, Lion 2020, Cheetah 2022; biosphere reserve zones; 18 BRs; Nilgiri first; MISHTI, Amrit Dharohar, Green Credit.
- **Waste and pollution (29, 30):** waste types; disposal methods; hierarchy; SWM Rules 2016; plastic (120 µm; SUP ban 2022); bio-medical colours; Basel, Rotterdam, Stockholm, Minamata; PM2.5, secondary pollutants,

smog, acid rain; AQI (6 categories), NCAP, GRAP; BOD, DO; Minamata (Hg), itai-itai (Cd), blue baby (nitrate), fluorosis; noise limits; Water Act 1974, Air Act 1981, EPA 1986.

EXAM TRAP

Most likely in the exam from this block: for **Group 2 Geography**, matching landforms with agents, local winds and ocean currents with places, layers of the Earth and the atmosphere, time from longitude, and India's climate types; for **Polity (Group 1 and Mains)**, the 73rd and 74th Amendments Article by Article and the committees; for **Economy**, national income terms and a calculation, PLFS terms, development indices and the Plans; for **Science and environment**, ecology terms, hotspots, conventions with their subjects, Indian wildlife laws and projects, waste rules and pollution diseases.

Revision test: 30 questions (60 minutes)

Answer all 30 without notes. Count: +1 for correct, −1/3 for wrong (APPSC style). Target: 24 or more.

Part A: Geography (1–10)

PRACTICE

1. Consider the following statements:
1. Venus and Uranus rotate from east to west.
2. A light year is a unit of time.
3. The Earth is at perihelion around 3 January.
4. A lunar eclipse can occur only on a full moon night.
Which of the above are correct?

A. 1 and 3 only

B. 1, 3 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. A light year is a unit of distance (about 9.46 trillion km).

PRACTICE

2. Match List-I with List-II:

List-I	List-II
A. Moho discontinuity	1. Boundary between crust and mantle
B. Asthenosphere	2. Reflects radio waves

C. Outer core	**3.** Soft upper mantle, source of magma
D. Ionosphere	**4.** Liquid layer that creates the Earth's magnetic field

(1) A → 1 B → 3 C → 4 D → 2

(2) A → 3 B → 1 C → 4 D → 2

(3) A → 1 B → 4 C → 3 D → 2

(4) A → 1 B → 3 C → 2 D → 4

Answer: (1) Moho: crust–mantle; asthenosphere: magma; outer core: magnetism; ionosphere: radio.

PRACTICE

3. Which one of the following pairs (original rock → metamorphic rock) is NOT correctly matched?

A. Granite → gneiss

B. Limestone → marble

C. Sandstone → quartzite

D. Shale → graphite

Answer: D. Shale becomes slate; coal becomes graphite.

PRACTICE

4. Assertion (A): The troposphere is thicker over the equator than over the poles.
Reason (R): Strong heating and convection at the equator carry air to greater heights.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Strong convection explains the greater height (about 18 km vs 8 km).

PRACTICE

5. Match List-I (landform) with List-II (agent):

List-I (landform)	List-II (agent)
A. Barchan	1. River in its old stage
B. Cirque	2. Groundwater in limestone
C. Stalagmite	3. Glacier
D. Oxbow lake	4. Wind

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) Barchan: wind; cirque: glacier; stalagmite: groundwater; oxbow lake: river.

PRACTICE

6. When it is 12 noon at Greenwich (0°), what is the local time at a place on 45° E longitude?

A. 9 a.m.

B. 3 p.m.

C. 6 p.m.

D. 2 p.m.

Answer: B. 45 × 4 min = 180 min = 3 hours ahead (east), so 3 p.m.

PRACTICE

7. Consider the following statements:

1. The Gulf Stream is a warm current.

2. The cold Benguela current is linked with the Namib desert.

3. The Kuroshio is a cold current.

4. El Niño years often bring a weak monsoon to India.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2, 3 and 4

D. 1, 2 and 4 only

Answer: D. The Kuroshio ('Japan current') is warm; the Oyashio is the cold one.

PRACTICE

8. Which one of the following statements is NOT correct?

A. The equatorial low-pressure belt is called the doldrums.

B. The polar regions are belts of low pressure.

C. The sub-tropical high-pressure belts are called the horse latitudes.

D. Sub-polar low-pressure belts lie at about 60°.

Answer: B. The poles have high pressure because the air there is very cold and dense.

PRACTICE

9. Assertion (A): The mesosphere is the coldest layer of the atmosphere.
Reason (R): The ozone in the mesosphere absorbs most of the Sun's ultraviolet radiation.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: the ozone layer is in the stratosphere, which is why the stratosphere warms with height.

PRACTICE

10. The Coromandel coast of Tamil Nadu gets most of its rain in winter from the north-east monsoon and has a dry summer. Which Köppen type fits it?

A. Amw

B. As

C. BWhw

D. Cwg

Answer: B. 's' means a dry summer; As is the tropical climate with summer drought and winter rain.

Part B: Polity and Economy (11–20)

PRACTICE

11. Consider the following statements about the 73rd Amendment:

1. Article 243A provides for the Gram Sabha.

2. The minimum age to contest panchayat elections is 21.

3. The Eleventh Schedule lists 18 subjects.

4. A State Finance Commission is set up every five years.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 4 only

D. 1, 2, 3 and 4

Answer: C. The Eleventh Schedule has 29 subjects; the Twelfth (municipalities) has 18.

PRACTICE

12. Match List-I (committee) with List-II (recommendation):

List-I (committee)	List-II (recommendation)
A. Balwant Rai Mehta	1. Three-tier Panchayati Raj
B. Ashok Mehta	2. Panchayats in Scheduled Areas (PESA)
C. L.M. Singhvi	3. Constitutional status for panchayats
D. Dileep Singh Bhuria	4. Two-tier system with mandal panchayats

(1) A → 4 B → 1 C → 3 D → 2

(2) A → 1 B → 4 C → 3 D → 2

(3) A → 1 B → 3 C → 4 D → 2

(4) A → 1 B → 4 C → 2 D → 3

Answer: (2) Balwant Rai Mehta: three tiers; Ashok Mehta: two tiers; Singhvi: constitutional status; Bhuria: PESA.

PRACTICE

13. Which one of the following statements about the 74th Amendment is NOT correct?

A. Wards Committees are needed where a municipality has three lakh or more people.

B. A Metropolitan Planning Committee is for areas of ten lakh or more people.

C. A Nagar Panchayat is for an area in transition from rural to urban.

D. The Twelfth Schedule lists 29 subjects.

Answer: D. The Twelfth Schedule has 18 subjects.

PRACTICE

14. A country's GDP at market prices is ₹500 crore, depreciation is ₹50 crore, net factor income from abroad is +₹10 crore and net indirect taxes are ₹40 crore. What is its national income (NNP at factor cost)?

A. ₹410 crore

B. ₹420 crore

C. ₹460 crore

D. ₹440 crore

Answer: B. 500 + 10 = 510; 510 − 50 = 460; 460 − 40 = 420.

PRACTICE

15. Assertion (A): Agriculture's share in India's employment is much larger than its share in GVA.
Reason (R): Output per worker in agriculture is low, and much of the farm workforce is disguisedly unemployed.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Low productivity and surplus labour explain why many workers produce a small share of output.

PRACTICE

16. Match List-I (index) with List-II (feature):

List-I (index)	List-II (feature)
A. PQLI	1. NITI Aayog's ranking of States
B. HDI	2. Deprivations in health, education and living standards (UNDP–OPHI)
C. MPI	3. UNDP index of health, education and income
D. SDG India Index	4. Morris D. Morris, 1979

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 1 D → 2

(4) A → 4 B → 3 C → 2 D → 1

Answer: (4) PQLI: Morris; HDI: UNDP; MPI: deprivations; SDG India Index: NITI.

PRACTICE

17. Arrange the following in chronological order:

1. Setting up of NITI Aayog

2. Setting up of the Planning Commission

3. Setting up of the National Development Council

4. Nationalisation of 14 major banks

A. 2, 3, 4, 1

B. 3, 2, 4, 1

C. 2, 4, 3, 1

D. 2, 3, 1, 4

Answer: A. 1950, 1952, 1969, 2015.

PRACTICE

18. Consider the following statements:

1. The Periodic Labour Force Survey is conducted by the NSO.

2. The unemployment rate is the unemployed as a share of the labour force.

3. Current Weekly Status uses a reference period of the last 7 days.

4. Usual status uses a reference period of the last 30 days.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. Usual status looks at the last 365 days.

PRACTICE

19. Which one of the following statements is NOT correct?

A. The Second Plan was based on the Mahalanobis model.

B. The Third Plan failed mainly because of wars and droughts.

C. The Harrod–Domar model says technology alone decides growth.

D. The term 'Hindu rate of growth' was coined by Raj Krishna.

Answer: C. Harrod–Domar stresses saving and the capital–output ratio; technology-led growth is Solow's idea.

PRACTICE

20. A district wants one body to combine the development plans of all its panchayats and municipalities into a single draft plan. Which constitutional body is meant for this?

A. State Finance Commission

B. District Planning Committee

C. Wards Committee

D. Gram Sabha

Answer: B. Article 243ZD provides a District Planning Committee to consolidate rural and urban plans.

Part C: Science and environment (21–30)

PRACTICE

21. Consider the following statements:

1. Energy flows in one direction through an ecosystem.

2. The pyramid of biomass in an ocean is usually inverted.

3. Lichens are pioneers of succession on bare rock.

4. The phosphorus cycle has a major gaseous phase.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. The phosphorus cycle is sedimentary; it has no significant gaseous phase.

PRACTICE

22. Match List-I (convention) with List-II (subject):

List-I (convention)	List-II (subject)
A. CITES	1. Wetlands
B. CMS (Bonn)	2. International trade in endangered species
C. Ramsar	3. Cross-border movement of hazardous waste
D. Basel	4. Migratory species

(1) A → 2 B → 4 C → 1 D → 3

(2) A → 4 B → 2 C → 1 D → 3

(3) A → 2 B → 1 C → 4 D → 3

(4) A → 2 B → 4 C → 3 D → 1

Answer: (1) CITES: trade; CMS: migratory species; Ramsar: wetlands; Basel: hazardous waste.

PRACTICE

23. Which one of the following statements is NOT correct?

A. India has parts of four of the world's 36 biodiversity hotspots.

B. The Eastern Ghats are one of the world's biodiversity hotspots.

C. Nilgiri was India's first biosphere reserve.

D. Nagarjunasagar–Srisailam is India's largest tiger reserve.

Answer: B. India's hotspots are the Himalaya, Indo-Burma, Western Ghats–Sri Lanka and Sundaland; the Eastern Ghats are not one.

PRACTICE

24. Assertion (A): Commercial international trade in Appendix I species of CITES is generally not allowed.
Reason (R): Appendix I lists species threatened with extinction.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Because these species face extinction, trade in them is banned.

PRACTICE

25. Match List-I (law) with List-II (feature):

List-I (law)	List-II (feature)
A. Forest Rights Act	1. 1972

B. Compensatory Afforestation Fund Act	**2.** 2002; National Biodiversity Authority
C. Biological Diversity Act	**3.** 2016; CAMPA
D. Wildlife (Protection) Act	**4.** 2006; Gram Sabha decides claims

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) FRA 2006; CAMPA Act 2016; BD Act 2002; WPA 1972.

PRACTICE

26. Consider the following statements:

1. Identified single-use plastic items were banned in India from 1 July 2022.

2. Plastic carry bags must be at least 120 microns thick from December 2022.

3. Sanitary landfills have liners to stop leachate.

4. Composting is an anaerobic process.

Which of the above are correct?

A. 1 and 3 only

B. 2, 3 and 4 only

C. 1, 2, 3 and 4

D. 1, 2 and 3 only

Answer: D. Composting is aerobic (needs air); biomethanation is the anaerobic process.

PRACTICE

27. Which one of the following pairs is NOT correctly matched?

A. IUCN Red List : a legally binding treaty

B. Project Tiger : 1973

C. Project Cheetah : Kuno National Park

D. MISHTI : mangrove planting

Answer: A. The IUCN Red List is a scientific assessment, not a law or treaty.

PRACTICE

28. Assertion (A): The Great Indian Bustard is listed as Critically Endangered.
Reason (R): Rollapadu in Andhra Pradesh is a sanctuary for the Great Indian Bustard.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but having a sanctuary does not explain why the bird is Critically Endangered (habitat loss and power lines do).

PRACTICE

29. Arrange the following in chronological order:
1. Project Cheetah
2. Project Tiger
3. Biological Diversity Act
4. Forest (Conservation) Act

A. 4, 2, 3, 1

B. 2, 3, 4, 1

C. 2, 4, 3, 1

D. 2, 4, 1, 3

Answer: C. 1973, 1980, 2002, 2022.

PRACTICE

30. In a district, many children have chalky, mottled teeth and some adults have stiff joints; tests show excess fluoride in borewell water. What is the condition?

A. Minamata disease

B. Itai-itai disease

C. Blue baby syndrome

D. Fluorosis

Answer: D. Excess fluoride causes dental and skeletal fluorosis (found in parts of Prakasam and Rayalaseema).

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

After the test: score = correct − (wrong ÷ 3). Write each wrong question's topic in the 'My mistakes' box below and re-read only that part of the day's notes. Revise this page again in 7 days.